



AmeraKey®

Controlled Quantum Key Distribution

Breaking The Shackles Of PKI

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Breaking the Shackles of PKI

Controlled Quantum Key Distribution (CQKD) — quantum-ready, purpose-built, for closed systems where the mission depends on the math, not the ceremony.

Public Key Infrastructure PKI is not broken; it's overextended. It was designed to let strangers verify each other across a hostile, public network — and at that job, it remains the foundation of the open Internet. The friction enterprises feel is not a failure of execution. The struggle between mixing these systems happens because we try to use solutions made for the public Internet to solve closed system challenges. Often doing our best to secure closed systems like factory floors, power stations, or hospital networks. These private areas need specialized protection that the public-trust model just wasn't built to handle. This friction between public and private environments often stems from a fundamental mismatch in operational logic: public systems rely on fixed endpoints and physical trust boundaries, while closed systems demand ephemeral scalability and decentralized identity. When long-lived infrastructure lifecycles collide with the rapid turnover of cloud-native certificates and inconsistent revocation lists, the resulting strain is why public key solutions in closed environments have historically fit like a square peg in a round hole.

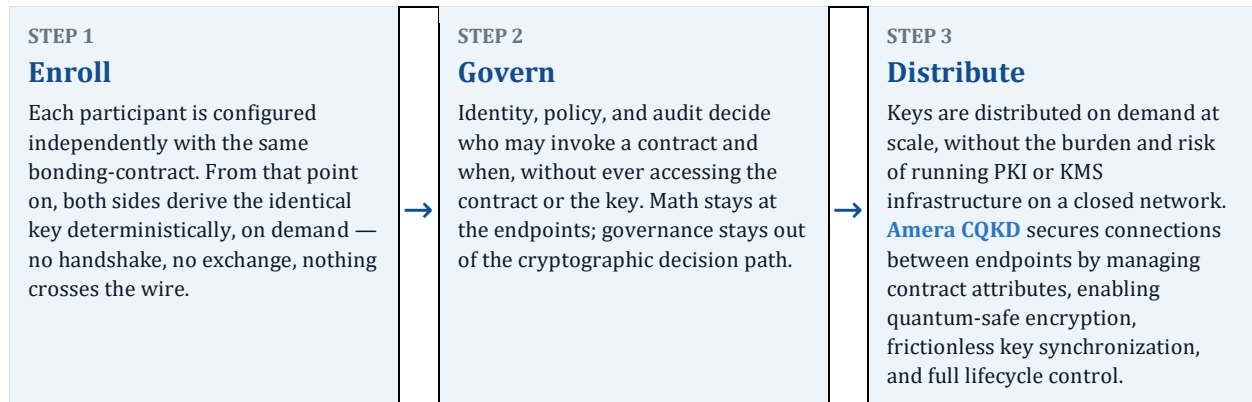
CQKD: A Different Model for a Different Problem

The AmeraKey Solution delivers Controlled Quantum Key Distribution — a deterministic, symmetric, certificate-free architecture for environments where trust is established physically and maintained mathematically. Two endpoints, bonded once during deployment, can independently and simultaneously generate identical encryption keys without ever transmitting key material across any wire. There is no certificate to expire, no authority to compromise, no revocation list to distribute, and no key escrow to defend.

Keys are derived on demand from a high-entropy image, a user-held credential, and a non-secret session factor issued by your control plane. Cryptographic work happens at the endpoints; the control plane sees only authorization, audit, and policy — never key material. The architecture is built on symmetric primitives suitable for post-quantum resilience while driving crypto-agility.

*"PKI is for public trust. **CQKD** is for controlled trust."*

How CQKD Works — Three Steps



Where CQKD Drives Value for Our Customers

- **OT, SCADA, and industrial control systems** where PLCs cannot be patched, certificates cannot be rotated without downtime, and CRL or OCSP infrastructure does not reach the substation reliably.
- **Defense and tactical enclaves** operating air-gapped or in degraded-comms environments where reliance on an online certificate authority creates operational friction or outright failure.
- **Private SaaS and multi-tenant data platforms** requiring per-tenant encryption isolation without the latency, cost, and exfiltration surface of a centralized key management service.
- **Long-lived IoT and embedded fleets** where 15-year asset lifecycles outlast every certificate, every CA root rotation, and every cryptographic standard PKI was originally built around.

CQKD vs. PKI — A Side-by-Side Look

Operational Phase	Traditional PKI	The Amera CQKD Solution
Enrollment	CSR, CA signing, chain validation, manual or scripted distribution.	Each endpoint independently configured with a bonding contract, nothing crosses the wire.
Key Distribution	Asymmetric wrap-and-ship; endpoints exchange or fetch certificates.	Keys distributed on demand at scale; same value materializes at every authorized endpoint.
Rotation	New certificate request, signing, redistribution, restart cycle.	Each invocation produces a fresh key — rotation is a property of use, not a separate operation.
Revocation	CRL or OCSP create operational friction in disconnected environments.	Authority to invoke the contract takes effect immediately, no CRL to push.
Recovery	Centralized key escrow or full re-enrollment.	Deterministic — the same contract reproduces the same keys, governed by deployment policy.

Operational Phase	Traditional PKI	The Amera CQKD Solution
Governance	Policy enforced via PKI infrastructure and audit overlays.	Identity, policy, and audit decide who may invoke a contract.
Quantum Posture	RSA and ECC are vulnerable to Shor's algorithm; full PQC migration required.	Symmetric primitives at PQC-appropriate key sizes quantum-safe by design, not by retrofit.
Storage Burden	Encrypted key blobs persisted in HSMs and KMS infrastructure.	No persisted private keys; contract attributes are inert outside an Amera -built runtime.
Closed-Network Fit	Designed for hostile public networks; assumes online CA and revocation services.	Designed for closed and disconnected systems; works air-gapped, intermittent, and at the edge.

What This Means for Your Organization

No certificate lifecycle to manage. No annual CA contract. No revocation infrastructure to monitor. No expirations to chase. The recurring spend on PKI tooling, audits, and incident response collapses into a one-time integration — and the integration doesn't depend on a certificate at all.

No keys in flight to intercept. Key material never crosses the wire. There are no persisted private keys to exfiltrate. There is no remote authority that can be compromised against you. The attack surface that PKI assumes — and tries to defend with layers of infrastructure — simply isn't there.

Already past Q-Day before it arrives. CNSA 2.0 and parallel national-security mandates are forcing migration to quantum-resistant algorithms on tight timelines. Harvest-Now-Decrypt-Later operations are already running against data with long-term value. With **CQKD** you're ahead of the procurement deadline, not chasing it.

Works when the network doesn't. Air-gapped, intermittently connected, geographically distributed, decade-lifetime — every property that strains PKI is the property **CQKD** was built for. No CA to reach. No CRL to fetch. No OCSP responder to fail. Your line stays up.

Schedule a CQKD Readiness Session In 45 minutes, we will map one of your controlled-trust environments — OT, SaaS tenant encryption, tactical systems, IoT fleet, or disconnected recovery — and identify where certificates, KMS calls, or stored key material can be removed. Bring one workflow. We will show where **CQKD** fits, where PKI should remain, and what a pilot would require.

Let PKI secure your Google searches.
Let **CQKD** secure your mission.